Strategic Lies and the Art of Truth Extraction

How to Design Rules That Make Honesty the Best Policy

Manas Nandan

open_board

Outline

1. The Problem of Lying
2. The Formal Framework
3. Auctions
4. Matching Markets
5. The Limits of Truthful Design
6. Conclusion

A Simple Model of a Lie: The Tax Evader

The Scenario

The Setup

A risk-neutral agent has true income Y and reports $\hat{Y} \in [0, Y]$.

- Tax rate: τ
- Audit probability: p
- Penalty multiplier: k

A Simple Model of a Lie: The Tax Evader

The Scenario

The Setup

A risk-neutral agent has true income Y and reports $\hat{Y} \in [0, Y]$.

- Tax rate: τ
- Audit probability: p
- Penalty multiplier: k

The Agent's Optimization Problem

Choose $\hat{Y}$ to maximize expected utility $E[U]$:

$$E[U(\hat{Y})] = \underbrace{(1-p)[\tau(Y-\hat{Y})]}_{\text{Gain if not caught}} + \underbrace{p[-k \cdot \tau(Y-\hat{Y})]}_{\text{Loss if caught}}$$

A Simple Model of a Lie: The Tax Evader

The Rational Choice

Factoring the expression for expected utility gives us:

$$E[U(\hat{Y})] = \tau(Y - \hat{Y}) \cdot [1 - p(1 + k)]$$

The agent's decision is driven entirely by the sign of the red term.

A Simple Model of a Lie: The Tax Evader

The Rational Choice

Factoring the expression for expected utility gives us:

$$E[U(\hat{Y})] = \tau(Y - \hat{Y}) \cdot [1 - p(1 + k)]$$

The agent's decision is driven entirely by the sign of the red term.

The Optimal Strategy

- If $1 - p(1 + k) > 0 \implies$ The payoff for lying is positive. A rational agent will lie as much as possible ($\hat{Y} \to 0$).
- If $1 - p(1 + k) < 0 \implies$ The payoff for lying is negative. A rational agent will tell the truth ($\hat{Y} \to Y$).

By designing the rules (p, k), we design the optimal action.

The Environment: A Formal Description

Any mechanism design problem can be described by a tuple:
$\mathcal{E} = (N, X, \Theta, \phi, (u_i)_{i \in N})$

- $N = \{1, \ldots, n\}$: A finite set of **agents**.
- X: A set of possible **outcomes**.
- $\Theta = \Theta_1 \times \cdots \times \Theta_n$: The **type space** (private information).
- $\phi(\theta)$: A **common prior belief** over the distribution of types.
- $u_i(x, \theta_i)$: Agent i's **utility function**.

The Environment: A Formal Description

Any mechanism design problem can be described by a tuple: $\mathcal{E} = (N, X, \Theta, \phi, (u_i)_{i \in N})$

- $N = \{1, \ldots, n\}$: A finite set of **agents**.
- X: A set of possible **outcomes**.
- $\Theta = \Theta_1 \times \cdots \times \Theta_n$: The **type space** (private information).
- $\phi(\theta)$: A **common prior belief** over the distribution of types.
- $u_i(x, \theta_i)$: Agent i's **utility function**.

A Critical Assumption: Quasi-Linear Utility

$$u_i(k, t_i, \theta_i) = v_i(k, \theta_i) - t_i$$

This form, where utility is (value - payment), is the bedrock of auction theory.

The Revelation Principle: A Huge Simplification

Theorem (Myerson, 1979)

Any outcome that can be achieved by any complex mechanism with rational agents can also be achieved by a simple, ***direct mechanism*** *where all agents find it optimal to report their private information truthfully.*

This allows us to stop designing complex games and instead focus on designing ***truthful outcome rules****.*

The Designer's Checklist (1/2): Incentive Compatibility

For a direct mechanism to be truthful, it must be **Incentive Compatible (IC)**.

Honesty Must Be the Best Policy

The expected utility from telling the truth (θ_i) must be at least as high as the utility from reporting any lie ($\hat{\theta}_i$).

The Designer's Checklist (1/2): Incentive Compatibility

For a direct mechanism to be truthful, it must be **Incentive Compatible (IC)**.

Honesty Must Be the Best Policy

The expected utility from telling the truth (θ_i) must be at least as high as the utility from reporting any lie ($\hat{\theta}_i$).

Bayesian IC

True on average, given beliefs about others.

$$E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta_i)] \geq E_{\theta_{-i}}[u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i)]$$

Dominant Strategy IC

True regardless of what others do. (The "gold standard").

$$u_i(f(\theta_i, \theta_{-i}), \theta_i) \geq u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i)$$

The Designer's Checklist (2/2): Individual Rationality

Agents will only play if they expect to benefit.

The Game Must Be Worth Playing

An agent's expected utility from participating must be at least their reservation utility (typically normalised to 0). This is the **Individual Rationality (IR)** constraint.

$$E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta_i)] \geq 0, \quad \forall i, \forall \theta_i$$

A mechanism is viable only if it satisfies both IC and IR.

First-Price Auction: The Bidder's Dilemma

The Setup

The Game

n bidders submit sealed bids. The highest bidder wins and pays their bid.

The Problem for a Bidder with Value v_i

They must choose a bid z to solve:

$$\max_z \quad \underbrace{(v_i - z)}_{\text{Profit if you win}} \cdot \underbrace{\text{Prob}(\text{Your bid } z \text{ is the highest})}_{\text{Chance of winning}}$$

- Bidding your true value ($z = v_i$) yields zero profit. You **must** lie.
- But how much should you "shade" your bid?

First-Price Auction: The BNE Derivation

Solving for the Rational Lie

We look for a symmetric equilibrium where everyone uses the same bidding function $b(v)$.

- The probability of winning with bid z is the chance that all $n-1$ other bidders have a value v_j such that their bid $b(v_j)$ is less than z.
- Assuming values $v \sim U[0,1]$, this gives:

$$\text{Prob(win)} = (b^{-1}(z))^{n-1}$$

- The bidder's problem is to choose a "fake" type v to report via the bid function $b(v)$:

$$\max_{v}(v_i - b(v))v^{n-1}$$

First-Price Auction: The BNE Derivation

Solving for the Rational Lie

We look for a symmetric equilibrium where everyone uses the same bidding function $b(v)$.

- The probability of winning with bid z is the chance that all $n-1$ other bidders have a value v_j such that their bid $b(v_j)$ is less than z.
- Assuming values $v \sim U[0,1]$, this gives:

$$\text{Prob(win)} = (b^{-1}(z))^{n-1}$$

- The bidder's problem is to choose a "fake" type v to report via the bid function $b(v)$:

$$\max_v (v_i - b(v))v^{n-1}$$

The Solution

Solving this maximization problem yields a differential equation whose solution is:

$$b(v_i) = \frac{n-1}{n} v_i$$

The Vickrey Auction: A Mechanism for Truth

A Small Change in Rules, A Huge Change in Strategy

The Vickrey (Second-Price) Rule

Highest bidder wins, but pays the **second-highest** bid.

Theorem

*In a Vickrey auction, bidding your true value ($b_i = v_i$) is a **dominant strategy**.*

The Intuition

Your bid determines *if* you win, but it does not determine the *price* you pay. This decouples the two strategic considerations. You can no longer gain by misrepresenting your willingness to pay.

Revenue Equivalence: A Surprising Result

Theorem (Revenue Equivalence Theorem)

Under standard assumptions (private values, risk-neutral bidders, etc.), any auction mechanism that ensures:

1. *The bidder with the **highest value** always wins the item.*
2. *A bidder with the **lowest possible value** expects zero profit.*

*will yield the **exact same expected revenue** for the seller.*

Revenue Equivalence: A Surprising Result

Theorem (Revenue Equivalence Theorem)

Under standard assumptions (private values, risk-neutral bidders, etc.), any auction mechanism that ensures:

1. *The bidder with the **highest value** always wins the item.*
2. *A bidder with the **lowest possible value** expects zero profit.*

*will yield the **exact same expected revenue** for the seller.*

What This Means

The aggressive bidding required in a **First-Price** auction is perfectly offset by the lower payment rule of a **Second-Price** auction. Strategically, they are very different, but on average, they deliver the same revenue to the seller.

Optimal Auctions: Beyond Revenue Equivalence

Myerson's Nobel-winning Insight

If all standard auctions are equivalent, how can a seller possibly increase their revenue?

By being willing to not sell the item.

Optimal Auctions: Beyond Revenue Equivalence

Myerson's Nobel-winning Insight

If all standard auctions are equivalent, how can a seller possibly increase their revenue?

By being willing to not sell the item.

The Core Idea

Instead of awarding the item to the highest bidder, the seller should award it to the bidder who provides the highest *net value* to the seller. This requires accounting for the "cost" of persuading bidders to reveal their high values, a cost captured by **virtual valuation**.

Virtual Valuation: What a Bid is Really Worth

Definition (Virtual Valuation)

For a bidder with value v drawn from a distribution with CDF $F(v)$ and PDF $f(v)$:

$$\varphi(v) = \underbrace{v}_{\text{True Value}} - \underbrace{\frac{1 - F(v)}{f(v)}}_{\text{Information Rent}}$$

The **information rent** is the discount the seller must give a high-value bidder to ensure they bid truthfully instead of pretending to have a lower value.

Virtual Valuation: A Regular Example

Example: Values are Uniform on $[0, 100]$

Here, $F(v) = v/100$ and $f(v) = 1/100$. This is a "regular" distribution.

- The information rent is $\frac{1-v/100}{1/100} = 100 - v$.
- The virtual valuation is $\varphi(v) = v - (100 - v) = \mathbf{2v - 100}$.
- A bid of \$80 has a virtual value of $2(80) - 100 = \$60$.
- A bid of \$40 has a virtual value of $2(40) - 100 = -\$20$.

The Optimal Auction (for Regular Distributions)

Myerson's Optimal Mechanism

To maximize revenue, the seller should:

1. For each bidder i with value v_i, calculate their virtual valuation $\varphi_i(v_i)$.
2. Allocate the item to the bidder j with the **highest non-negative** virtual valuation.
3. If all virtual valuations are negative, the seller **keeps the item**.

The Optimal Auction (for Regular Distributions)

Myerson's Optimal Mechanism

To maximize revenue, the seller should:

1. For each bidder i with value v_i, calculate their virtual valuation $\varphi_i(v_i)$.
2. Allocate the item to the bidder j with the **highest non-negative** virtual valuation.
3. If all virtual valuations are negative, the seller **keeps the item**.

Applying to our Regular Example (Uniform on $[0, 100]$)

The rule is to sell to the bidder with the highest $\varphi(v) = 2v - 100$, as long as it's positive.

- A virtual value is positive only if $2v - 100 > 0 \implies v > \$50$.
- This mechanism is equivalent to running a standard Second-Price auction with a **reserve price of \$50**.

The Seller's True Goal: Maximizing Revenue

Beyond just selling to the highest bidder

The Core Principle of Optimal Auctions

A seller focused purely on maximizing revenue should not care about a bidder's true value (v), but rather the **marginal revenue** that bidder generates.
In auction theory, this marginal revenue is captured by the **virtual valuation** ($\varphi(v)$).

The Optimal Rule

To maximize revenue, the seller must follow a simple two-part rule:

1. Only sell to a bidder if their marginal revenue (virtual value) is non-negative.
2. If multiple bidders meet this condition, sell to the one with the **highest** marginal revenue.

The Conflict in a Non-Regular World

When a higher value is not a better deal

Comparing Two Bidders

Let's compare two bidders from the "problematic" range of this distribution:

- **Bidder 1:** Has a true value of $v_1 = 0.01$.
  - Their marginal revenue is $\varphi(v_1) = -\mathbf{0.17}$.
- **Bidder 2:** Has a higher true value of $v_2 = 0.04$.
  - Their marginal revenue is $\varphi(v_2) = -\mathbf{0.28}$.

The Stable Marriage Problem

e.g., Matching Men $\leftrightarrow$ Women

The Goal: Stability

The goal is to arrange marriages so that no two people who are not married to each other would both rather be with each other than their current partners.

The Stable Marriage Problem

e.g., Matching Men $\leftrightarrow$ Women

The Goal: Stability

The goal is to arrange marriages so that no two people who are not married to each other would both rather be with each other than their current partners.

Definition (Blocking Pair)

A pair (m, w) is a **blocking pair** if man m and woman w are not married to each other, but both would prefer each other over their assigned partners. A matching is **stable** if no such pairs exist.

The Gale-Shapley Algorithm (1962)

A simple, iterative algorithm that finds a stable matching.

The Algorithm (Men-Proposing Version)

1. **Round 1:** Every man proposes to his #1 ranked woman.
2. Women review their proposals, "tentatively hold" their favorite suitor, and reject the rest.
3. **Subsequent Rounds:** Each rejected man proposes to his next choice. Women again compare new proposals to their tentative partner, keeping the one they prefer.
4. **Termination:** The algorithm ends when no man is rejected.

The Power of the Gale-Shapley Algorithm

Key Properties

- **Guaranteed Stability:** The algorithm is mathematically proven to always produce a stable matching. It systematically eliminates all potential blocking pairs.
- **Optimal for Proposers:** The resulting matching is universally better for the proposing side (men, in our example) than any other stable matching.
- **Strategy-Proofness:** It is a **dominant strategy** for the proposing side to report their preferences truthfully. Misrepresenting preferences cannot get them a better partner and can easily result in a worse one.

This work was a key part of the 2012 Nobel Prize in Economic Sciences.

The Fundamental Limit: General Voting

Is it always a good idea to design a strategy-proof mechanism?

The Fundamental Limit: General Voting

Is it always a good idea to design a strategy-proof mechanism?

The Setting of General Voting

- We have 3 or more alternatives (candidates).
- Voters can have *any* possible preference ranking over them.
- The voting rule must be non-trivial (any candidate can win).

The Gibbard-Satterthwaite Impossibility Theorem

Theorem (Gibbard-Satterthwaite 1973, 1975)

For three or more alternatives, any voting rule that is:

1. ***Strategy-Proof*** *(truth-telling is always optimal)*
2. ***Non-Trivial*** *(not just a fixed outcome)*

must be a ***dictatorship****.*

Sobering Implication

The Takeaway

For any reasonable, non-dictatorial voting system, there will always be a situation where a voter has a rational incentive to lie about their preferences (i.e., vote strategically).

Why did auctions and matching work?

They operated on **restricted domains**. By assuming quasi-linear utility, for example, we did not allow agents to have *any* possible preference structure. This restriction is what allows us to escape the impossibility result.

Conclusion: Truth is Designed, Not Discovered

- Rational people lie when the rules of the game reward them for it.
- We cannot force people to be honest, but we can design the rules to make honesty the optimal strategy. This is the art of truth extraction.
- This requires careful mathematical and economic engineering, from the tax code to spectrum auctions to kidney exchanges.
- Ultimately, truth in a strategic society is not a pre-existing thing we simply discover.

It is constructed.

Q&A and Further Reading

Thank You

Questions?

- **Hurwicz (1972)**: "On Informationally Decentralized Systems." Crux: Laid the foundations of mechanism design. *Nobel 2007 (with Maskin, Myerson).*
- **Maskin (1999)**: "Nash Equilibrium and Welfare Optimality." Crux: Developed theory of implementation. *Nobel 2007.*
- **Myerson (1979)**: "Incentive Compatibility and the Bargaining Problem." Crux: Revelation Principle in Bayesian settings. *Nobel 2007.*
- **Myerson (1981)**: "Optimal Auction Design." Crux: Characterized revenue-maximizing auctions using virtual valuations.
- **Vickrey (1961)**: "Counterspeculation, Auctions, and Competitive Sealed Tenders." Crux: Introduced second-price auction. *Nobel 1996.*
- **Gale & Shapley (1962)**: "College Admissions and the Stability of Marriage." Crux: Deferred Acceptance algorithm. *Nobel 2012 to Shapley.*
- **Roth (1984)**: "Labor Market for Medical Interns." Crux: Applied stable matching to real markets. *Nobel 2012.*
- **Roth et al. (2004)**: "Kidney Exchange." Crux: Practical mechanisms for complex barter markets. *Nobel 2012 contribution.*